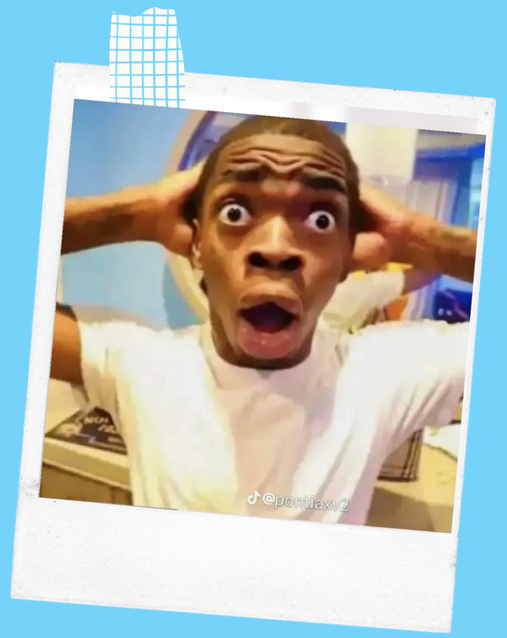
CADET

Decision Support System

BACKGROUND

Developed as a tool to help clinicians improve early cancer detection by analyzing patient data patterns. The development of CaDet was driven by the limitations of current approaches in cancer screening, such as human factors and complex data evaluation.



PROBLEM ADDRESSED

The problem that CaDet addressed is the limitation of cancer risk evaluation and early detection.

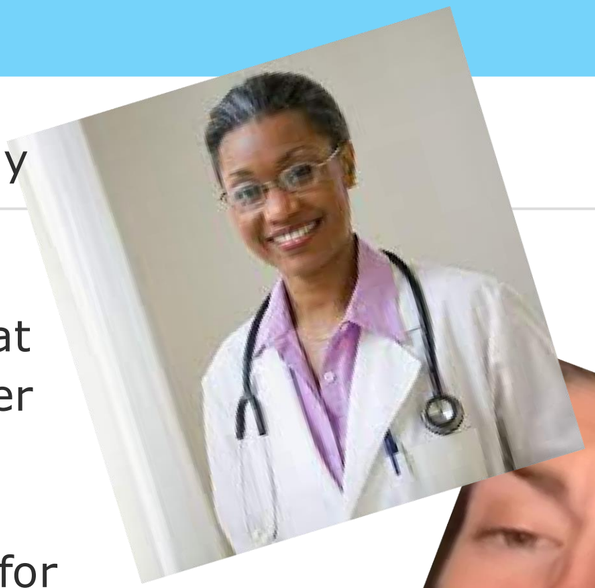
Early detection is critical in improving cancer outcomes, as cancers that are detected at an early stage are more likely to be treated successfully. However, it can be challenging, as cancer can be asymptomatic in its early stages, and symptoms may not appear until the disease has progressed.

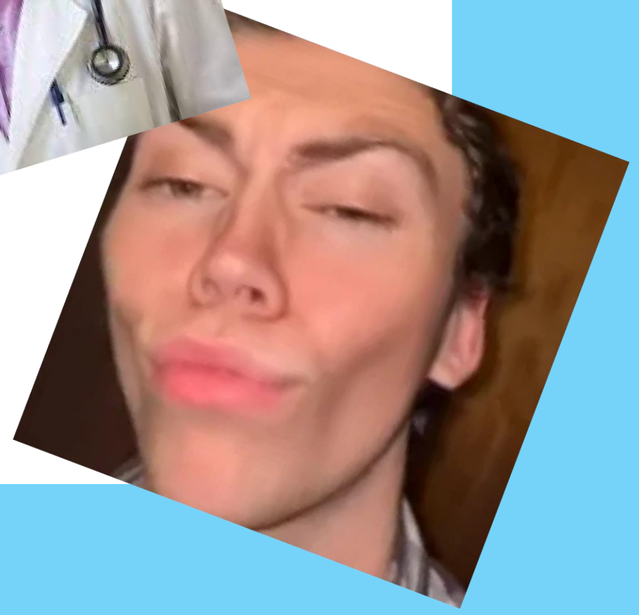
EXPERTS IT MIMICS

The CaDet expert System mimics a physician, specifically their job in early cancer detection.

The program presents physicians with data patterns that may require clinical attention with regard to early cancer detection.

Instead of the physicians, the program does it for them for faster the diagnosis.





WHAT ARE THE ADVANTAGES OF THE EXPERT SYSTEM?

1. Can detect cancer earlier.
2. Data necessary for analysis is easy to gather. (e.g. epidemiologic data)
3. It can detect cancer through common means like questionnaires and common medical tests.

WHAT ARE THE CHALLENGES?

1. Data quality - Depends on the quality used to train them and requires a large and diverse dataset that is representative of population being screened.

2. Model accuracy- False positives or false negatives can have serious consequences for patients

3. Ethical concern - the use of CaDet raises ethical concerns, such as ensuring patient privacy and informed consent.

4. Integration with existing systems - Requiring careful planning and coordination is required to ensure compatibility and minimize disruption to existing workflows.

References:

- Fuchs, J., Heller, I., Topilsky, M., & Inbar, M. (1999). CaDet, a Computer-Based Clinical Decision Support System for Early Cancer Detection. Cancer Detection and Prevention, 23(1), 78–87. https://doi.org/10.1046/j.1525-1500.1999.09902.x
- HARSHIL PATEL. (2020, July 25). CADET AI IA [Video]. YouTube. https://www.youtube.com/watch?v=SpDD29-qOZE
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